

Supplementary Note: The q_2 Half-Life MPL-LD Residual Response

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1 Purpose

This note records the current theoretical interpretation of the residual correction used in the main paper. It replaces the older DropRelaxS-style supplement. The goal is not to prove a universal optimizer law, but to make explicit why the current estimator has the form

$$\hat{L}_s(t) = L_{\text{MPL},s}(t) + a_s \hat{\kappa}_s \phi_{\lambda_s,s}(t). \quad (1)$$

The three structural choices are:

1. a causal LR-drop response feature $\phi_{\lambda_s,s}$;
2. a schedule-only q_2 half-life rule for λ_s ;
3. MPL-LD tangent projection before estimating the single scalar $\hat{\kappa}_s$.

2 Linear Response Approximation

Let e_t be an unobserved excess-loss tracking state after the MPL baseline has modeled the dominant quasi-static trend. Around a fixed source schedule, a first-order local model is

$$e_t = A_t e_{t-1} + b d_t + \epsilon_t, \quad d_t = [\eta_{t-1} - \eta_t]_+. \quad (2)$$

The positive LR decrement d_t is the forcing term: when the LR drops, the ideal LR-dependent loss floor moves, while the observed training state responds with finite memory. For a stable one-pole approximation in cumulative-LR time,

$$A_t \approx \exp(-\lambda_s \eta_t). \quad (3)$$

Substituting this into Eq. (2) gives

$$z_t = \exp(-\lambda_s \eta_t) z_{t-1} + d_t, \quad \phi_{\lambda_s,s}(t) = z_t / \eta_{\max}. \quad (4)$$

Equivalently,

$$\phi_{\lambda_s,s}(t) = \frac{1}{\eta_{\max}} \sum_{u \leq t} d_u \exp\left(-\lambda_s \sum_{v=u+1}^t \eta_v\right). \quad (5)$$

This derivation explains the qualitative shape of the feature: it is causal, drop-only, and measured in cumulative LR. The scalar κ absorbs the unknown mapping from the hidden state to loss units.

3 q_2 Half-Life Rule

The response rate is not fitted from WSD targets. It is computed from the target LR schedule. For $D_s = \sum_t d_t$, define

$$p_t = \frac{d_t}{D_s}, \quad q_s = \sum_t p_t^2 = \frac{1}{n_{\text{eff}}}. \quad (6)$$

Thus q_s is the inverse effective number of drop atoms. A single sharp drop has $q_s = 1$; a drop uniformly spread over n steps has $q_s = 1/n$.

The observed curves are logged mainly every $\Delta_{\text{obs}} = 128$ steps. We set

$$\lambda_{\text{obs}} = \frac{\log 2}{\eta_{\text{max}} \Delta_{\text{obs}}}, \quad (7)$$

and use the half-life bracket

$$H_s = (2 - q_s) \Delta_{\text{obs}}, \quad \lambda_s = \frac{\log 2}{\eta_{\text{max}} H_s} = \frac{\lambda_{\text{obs}}}{2 - q_s}. \quad (8)$$

The interpretation is simple: diffuse decay is assigned roughly a two-observation half-life, while a single-step drop is assigned a one-observation half-life. This is a resolution-based structural prior, not a fitted physical constant.

4 Support-Projection Locality

The correction should represent local cooldown forcing, not full-horizon diffuse decay. Let the post-warmup horizon be $H = T_s - W$, and let ℓ_s be the positive-drop support span. Let m_s be the uniform density on that support, and let u be the uniform density on the full horizon. Projecting away the diffuse mode gives

$$a_s = \mathbf{1}\{D_s > 0\} \frac{\|(I - P_u)m_s\|_2^2}{\|m_s\|_2^2}. \quad (9)$$

Since $\|m_s\|_2^2 = 1/\ell_s$ and $\|P_u m_s\|_2^2 = 1/H$,

$$a_s = \mathbf{1}\{D_s > 0\} \left[1 - \frac{\ell_s}{T_s - W} \right]_+. \quad (10)$$

This is why the term is not a learned gate. It is an explicit energy fraction after removing the global diffuse component.

5 MPL-LD Nuisance Projection

The source residual is

$$r_{\text{cos}}(t) = L_{\text{cos}}(t) - L_{\text{MPL,cos}}(t). \quad (11)$$

If the LR-dependent MPL parameters are locally perturbed, the residual contains

$$J_{\text{LD}}(t) \Delta \theta_{\text{LD}}, \quad J_{\text{LD}}(t) = \left[\frac{\partial L_{\text{MPL}}}{\partial \log B}, \frac{\partial L_{\text{MPL}}}{\partial \log C}, \frac{\partial L_{\text{MPL}}}{\partial \log \beta}, \frac{\partial L_{\text{MPL}}}{\partial \log \gamma} \right]. \quad (12)$$

Let P_{LD} be the orthogonal projection onto this four-dimensional tangent space on the cosine calibration suffix. For target schedule s ,

$$x_s = (I - P_{\text{LD}}) \phi_{\lambda_s, \text{cos}}, \quad y = (I - P_{\text{LD}}) r_{\text{cos}}. \quad (13)$$

The fitted amplitude is

$$\hat{\kappa}_s = \frac{\langle x_s, y \rangle_+}{\|x_s\|_2^2 + 1/N_{\text{cal}}}. \quad (14)$$

Only this scalar is estimated from loss residuals. The response rate, locality factor, and nuisance subspace are all determined by the LR schedule and frozen MPL formula.

6 Current Evidence and Boundaries

The current theory-refinement audit reports:

split	mean MAE change	worst row	wins
same-scale WSD-family	−29.88%	−4.67%	15/15
cross-scale WSD-family	−24.95%	−3.15%	30/30
controls	+0.00%	+0.00%	9/9 non-harm

The main negative controls are equally important. Without MPL-LD nuisance projection, the raw one-dimensional projection fails on WSD-family targets. Without support-projection locality, the WSD mean improves slightly but the short-cosine control is harmed. Density projection is natural but still keeps too much diffuse cosine signal. These failures justify the current low-capacity formula and also define its limitations.